

# ACADEMIC TASK – 1 (CA1)

**COURSE:** ENVIRONMENTAL STUDIES

**COURSE CODE:** CHE110

TOPIC

## SUSTAINABLE SUPPLY CHAIN MANAGEMENT PRACTICES

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# 1. INTRODUCTION

A supply chain is the complete system through which raw materials, products, information and services move from suppliers to manufacturers, distributors and finally to customers.

In the modern business environment, companies are not only concerned about reducing cost and increasing profit. They also need to consider the impact of their activities on the environment and society. This has created the need for **Sustainable Supply Chain Management (SSCM)**.

Sustainable Supply Chain Management means managing supply-chain activities in a way that balances **economic performance, environmental protection and social responsibility**.

It includes sustainable purchasing, responsible sourcing of raw materials, energy-efficient production, reduction of waste, environmentally friendly transportation, proper packaging and responsible disposal of products.

## 2. MEANING OF SUSTAINABLE SUPPLY CHAIN MANAGEMENT

Sustainable Supply Chain Management is an approach in which a company manages its suppliers, production, transportation, distribution and product disposal while considering environmental, social and economic responsibilities.

It aims to reduce environmental damage and resource consumption without affecting the long-term profitability and competitiveness of the business.

## 3. OBJECTIVES

1. Reduce environmental pollution caused by business activities.
2. Reduce carbon emissions from production and transportation.
3. Use natural resources efficiently and avoid unnecessary consumption.
4. Reduce, reuse and recycle waste generated throughout the supply chain.
5. Promote ethical sourcing of raw materials.
6. Improve working conditions for employees and workers.
7. Reduce unnecessary packaging and encourage eco-friendly materials.
8. Improve long-term business performance through efficient resource use.
9. Encourage suppliers to follow sustainable practices.
10. Meet the expectations of environmentally conscious consumers.

## 4. THREE DIMENSIONS OF SUSTAINABLE SUPPLY CHAIN

**Environmental Dimension:** Focuses on reducing carbon emissions, saving water and energy, using renewable energy, reducing waste and recycling materials.

**Social Dimension:** Focuses on safe working conditions, fair wages, employee welfare, equal opportunities and respect for human rights.

**Economic Dimension:** Focuses on reducing operating costs, improving efficiency, reducing material wastage, improving quality and maintaining long-term profitability.

**Sustainability Triangle:** Environmental + Social + Economic = Sustainable Supply Chain.

## 5. MAJOR SUSTAINABLE SUPPLY CHAIN MANAGEMENT PRACTICES

- 1. Sustainable Procurement:** Selecting suppliers that use environmentally responsible and ethical methods.
- 2. Green Manufacturing:** Reducing energy consumption, water usage, emissions and production waste.
- 3. Sustainable Transportation:** Using fuel-efficient or electric vehicles, route optimisation and consolidated shipments.
- 4. Eco-Friendly Packaging:** Reducing plastic and using recyclable, reusable or biodegradable materials.
- 5. Waste Management:** Reducing waste at the source and reusing or recycling materials.
- 6. Renewable Energy:** Using solar, wind and other renewable sources in factories, warehouses and offices.
- 7. Reverse Logistics:** Moving products or materials back for repair, reuse, recycling or proper disposal.
- 8. Supplier Sustainability Assessment:** Evaluating suppliers on environmental performance, labour practices and ethical standards.

## 6. IMPORTANCE OF SUSTAINABLE SUPPLY CHAIN MANAGEMENT

Sustainable supply-chain management is important because supply-chain activities can have significant environmental, social and economic effects.

**Environmental benefits:** Reduction in greenhouse-gas emissions, less waste, better use of resources, lower water and energy consumption, reduced pollution and increased recycling.

**Social benefits:** Better working conditions, protection of worker rights, ethical treatment of employees and support for local communities.

**Economic benefits:** Lower energy and material costs, better operational efficiency, reduced waste-related expenses, improved brand reputation and long-term business resilience.

## 7. CHALLENGES OF SUSTAINABLE SUPPLY CHAIN MANAGEMENT

- 1. High Initial Investment:** Cleaner technology and renewable energy can require significant investment.
- 2. Lack of Supplier Cooperation:** Some suppliers may lack technology, knowledge or resources.
- 3. Difficulty in Monitoring:** Large companies may have many suppliers to monitor.
- 4. Higher Short-Term Costs:** Sustainable materials and technologies can sometimes cost more initially.
- 5. Lack of Awareness:** Small businesses may lack sustainability knowledge.
- 6. Transportation Challenges:** Long-distance transportation can increase fuel use and emissions.
- 7. Measuring Sustainability:** It can be difficult to measure the complete environmental and social impact.
- 8. Balancing Profit and Sustainability:** Businesses may need to balance short-term financial goals with long-term sustainability.

## 8. CASE STUDY – IKEA AND SUSTAINABLE SUPPLY CHAINS

IKEA is a global home-furnishing company that provides an example of how sustainability can be incorporated into business and supply-chain activities.

The company has focused on responsible sourcing, renewable energy, resource efficiency, sustainable materials and circular business practices.

**Sustainable Sourcing:** IKEA works with suppliers and promotes responsible sourcing of materials.

**Renewable Energy:** The company has invested in renewable energy and energy-efficiency initiatives.

**Sustainable Materials:** It has increased its focus on renewable and recycled materials in products and packaging.

**Waste Reduction:** Efficient product design and material use can help reduce waste.

**Circular Economy:** IKEA has explored ways to extend product life through reuse, repair, resale and recycling.

The case shows that sustainability can be considered across sourcing, production, packaging, transportation, customer use and recycling.

## 9. PROPOSED PRODUCT – GREENCHAIN

**Product Name:** GreenChain – Sustainable Supply Chain Tracker

GreenChain is a simple digital tool designed to help small businesses monitor the sustainability of their supply-chain activities.

**It can track:** supplier sustainability, energy consumption, transportation, packaging materials, waste generation and recycling.

**Main features:**

1. Supplier sustainability checklist
2. Waste tracking
3. Packaging assessment
4. Transportation monitoring
5. Monthly sustainability score
6. Suggestions for reducing environmental impact

A business can give scores from 1 to 5 for Supplier, Production, Transport, Packaging and Waste. The scores can be combined to create an overall sustainability score.

## 10. KEY FINDINGS

1. Sustainability needs to be considered throughout the entire supply chain.
2. Sustainable procurement can influence supplier environmental performance.
3. Transportation is an important area for reducing fuel use and emissions.
4. Eco-friendly packaging can reduce unnecessary waste.
5. Recycling and reverse logistics can support circular use of resources.
6. Sustainable practices can provide environmental and economic benefits.
7. Supplier cooperation is essential for supply-chain sustainability.
8. Technology can help businesses monitor and improve sustainability performance.
9. Sustainable supply chains can improve customer trust and business reputation.
10. Long-term sustainability requires cooperation among businesses, suppliers, employees, customers and governments.

## 11. RECOMMENDATIONS

1. Select suppliers using clear sustainability criteria.
2. Reduce unnecessary packaging.
3. Increase the use of recycled and renewable materials.
4. Use energy-efficient machinery and equipment.
5. Increase the use of renewable energy.
6. Improve transportation and delivery routes.

7. Encourage recycling and reuse.
8. Train employees and suppliers about sustainability.
9. Regularly measure environmental performance.
10. Use technology to track sustainability indicators.
11. Develop long-term relationships with responsible suppliers.
12. Consider environmental and social impacts while making business decisions.

## 12. CONCLUSION

Sustainable Supply Chain Management is an important approach for modern businesses because it connects business performance with environmental and social responsibility.

A sustainable supply chain considers the complete process, starting from sourcing raw materials and continuing through production, transportation, packaging, distribution, customer use and disposal or recycling.

Practices such as sustainable procurement, green manufacturing, renewable energy, eco-friendly packaging, waste reduction, sustainable transportation and reverse logistics can help businesses reduce their environmental impact.

Although companies may face challenges such as initial investment, supplier cooperation and monitoring difficulties, sustainable supply-chain practices can provide long-term benefits through better efficiency, reduced waste, improved reputation and responsible resource use.

Therefore, sustainable supply chain management can help businesses achieve a balance between **profit, people and the planet**. It is an important strategy for building resilient and successful businesses in the future.

## 13. REFERENCES

1. Environmental Studies – Course material, CHE110.
2. United Nations – Sustainable Development Goals.
3. International Labour Organization – Sustainable and responsible business practices.
4. IKEA – Sustainability and responsible sourcing information.
5. World Business Council for Sustainable Development – Sustainable business resources.

**DEMONSTRATION VIDEO LINK :-** <https://www.instagram.com/p/DdO-uBgPlpe/?hl=en>